

Tzu-Han Lin

Elizabeth, NJ | <https://tzuhan2424.github.io/> | lintzuh@kean.edu | <https://www.linkedin.com/in/tzuhanlin/>

EXPERIENCE

Software Developer Kean University	Sep 2023 – May 2024
– Developed and deployed a full-stack ASP.NET system for IPEDS reporting on microsoft IIS server, streamlining data validation and minimizing redundant interdepartmental email communications by 20%. – Developed a robust front-end validation system with ReactJS, enhancing user experience through datatype checks, input range validation, and intuitive tooltips, resulting in a 40% reduction in user input errors across the platform. – Designed and implemented APIs using C# to optimize back-end processes, including server-side validation and database interactions, improving system reliability.	
Programming Tutor Kean University	Sep 2022 – May 2023
– Mentored 10 students weekly, assisting with MySQL, PHP, data structures, Java, and Python, resulting in a 20-40% improvement in their grades.	

AWARDS

Dean List Kean University	Dec 2022
Summer Research Grant: Students Partnering with Faculty (SpF) Kean University	May 2022, 2023

TECHNICAL SKILLS

Programming Languages : C, C#, Python, php, Java ,JavaScript, HTML/CSS
Databases : MS-SQL, MySQL,PL/SQL, MongoDB
Web Frameworks : ASP.NET (.NET Core), React, Flask, Django, Express.js
DevOps and API Tools : Git, Docker, Jira
Cloud and Server configuration : Azure Cloud, Microsoft IIS, Linux
Deep Learning frameworks : Pytorch

EDUCATION

Kean University	Master in Computer Information Systems GPA : 4.0	Jan 2022 – May 2024
National Tsing Hua University	Bachelor of Science in Medical Science	Sep 2013 – Jun 2017

RESEARCH

Selective Image-Based Weakly Supervised Segmentation in Breast Ultrasound	May 2023 – Jan 2024
– Accepted by the 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2024 – Established a weakly supervised image segmentation framework, integrating an innovative auto-selection algorithm to identify high-potential candidates from class activation maps, resulting in an 8% increase in Intersection over Union (IoU) score compared to non-selected training set.	

PROJECTS

TideTrack: Harmful algal bloom observing and forecasting system React, Python, Django, MySQL, Azure	Jan-Apr 2024
– Launched and deployed TideTrack on Azure Cloud, a system designed to perform ETL (Extract, Transform, Load) operations on diverse data sources (NOAA) and forecast the intensity of harmful algal blooms in the Gulf of Mexico. – Developed a React-based frontend integrated with ArcGIS API to visualize harmful algae areas on a map, providing users with real-time data reports for informed decision-making. – Developed a backend using Django as the framework to fetch data from an Azure MySQL cloud database, allowing users to search within specific date ranges and apply custom conditions.	
The Virtual Try-on Booth Flask, Python, Docker	Sep-Dec 2023
– Designed and constructed backend APIs for an e-commerce clothing website, integrating multiple deep learning models to enable an image-based virtual try-on feature, resulting in a 30% increase in user engagement	
Inventory Management and Order Processing System HTML/CSS, PHP, MySQL, Linux	Sep-Dec 2023
– Launched an online inventory and order management system utilizing concurrent transactions to prevent conflicts and ensure data consistency across multiple user interactions, improving system reliability – Implemented database triggers to enhance auditing capabilities, ensuring robust monitoring and historical tracking of table modifications for improved operational accountability.	
Mars Terrain Type Segmentation with Novel Edge Detection Python, PyTorch	Jan-May 2023
– Performed comparative analysis between a Fully Convolutional Network (FCN) and a custom FCN with edge feature channel for image segmentation, focusing on Mars terrain type. – Improved the bedrock type recall rate by 5%.	